

8D PROBLEM SOLVING REPORT

Subject: Recurring Leakage on Centrifugal Pump P-xxx (Mechanical Seal)

Report No.:	
Date Opened:	
Date Closed:	
Prepared By:	
Customer / Process Affected:	Cooling Water Circulation Loop - Utilities Area
Equipment / Part No.:	Mechanical Seal

D0 — Plan — Emergency Response / Symptom Recognition

Operator on routine rounds observed continuous drip leakage (~2-3 drops/sec) from the mechanical seal area of Pump P-xxx on 2025-xx-xx. Leakage rate increased over the following 24 hours to a steady stream. Standby pump P-xxx was started and P-xxx was tagged out and isolated pending investigation.

D1 — Establish the Team

Name	Role	Department	Responsibility
	Team Leader	Reliability	Coordinate investigation, own 8D closure
	Maintenance Lead	Maintenance	Teardown, inspection, seal replacement
	Process Engineer	Operations	Operating condition review, cavitation check
	Quality Engineer	Quality	Root cause verification, CAPA tracking
	Seal Technical Support	External - Seal OEM	Seal selection & failure mode review

D2 — Describe the Problem (5W2H)

What:	Mechanical seal leakage from Pump P-xxx exceeding acceptable drip rate (>1 drop/10 sec per site standard)
Where:	Seal chamber / gland area of P-xxx, Cooling Water Circulation Loop, Utilities Area
When:	First observed 2025-xx-xx, worsened progressively over ~24 hrs
Who Found It:	Operations - routine rounds (Shift B)
How Much / Extent:	~15 L/hr at peak leakage before shutdown; 3rd recurrence in 8 months on same pump
How Detected:	Visual observation during rounds; confirmed by vibration spot-check and seal chamber pressure log

D3 — Interim Containment Actions (ICA)

Containment Action	Owner	Date	Status
Isolate P-xxx, switch cooling duty to standby P-xxx	Operations		Complete
Contain leaked fluid, clean spill area, inspect for slip hazard	Operations		Complete
Increase inspection rounds on P-xxx to 2x/shift until root cause closed	Operations		Complete

D4 — Root Cause & Escape Point Analysis

Detailed 5-Why analysis performed for the two contributing failure branches (seal face wear and installation/alignment) — see tab '5 Why Analysis'.	
Root Cause (Primary):	Pump repeatedly operated below Minimum Continuous Stable Flow (MCSF) during low cooling-demand periods, causing recirculation-induced cavitation that eroded the mechanical seal faces and caused premature seal failure.

D5 — Choose & Verify Permanent Corrective Actions (PCA)

Full corrective and preventive action plan, owners, dates, and verification method are tracked in tab 'CAPA Action Plan'. Summary of selected permanent corrective actions below.		
Corrective Action	Type	Verification Method
Install low-flow switch/alarm on P-xxx discharge, interlocked to alert Operations below MCSF	Corrective	Functional test of interlock at simulated low-flow; signed off by Instrumentation
Replace damaged seal with OEM-recommended upgraded seal face material (SiC/SiC) rated for the duty	Corrective	Post-installation vibration & leak inspection at 24h, 7d, 30d run

Correct shaft/coupling alignment to OEM tolerance during reinstallation	Corrective	Laser alignment report filed with work order
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D6 — Implement & Validate Permanent Corrective Actions

Seal replaced and low-flow alarm commissioned on 2025-xx-xx. P-xxx returned to service 2025-xx-xx. Monitored for 6 days with zero leakage, seal chamber pressure and vibration within normal band, and no low-flow alarm activations under normal demand swings. Standby P-xxx kept available during the validation window.

D7 — Prevent Recurrence (Systemic / Preventive Actions)

Preventive Action	Applies To	Owner	Target Date
Add minimum-flow interlocks to all critical centrifugal pumps per updated design standard	All critical pumps	Reliability Eng.	01/10/2025
Update PM plan to include quarterly seal chamber pressure & vibration	P-xxx & similar pumps	Maintenance Planner	01/09/2025
Add MCSF operating window to pump datasheets and control room reference cards	All centrifugal pumps	Process Engineering	15/09/2025
Train operations on low-flow risk and cavitation symptoms	Operations Shift Teams	Training Coordinator	15/09/2025
Update FMEA for cooling water pumps to capture off-BEP operation	Cooling Water System FMEA	Quality	15/10/2025

D8 — Congratulate the Team / Closure

Root cause identified and eliminated; permanent corrective actions verified effective over the validation window; systemic preventive actions assigned with target dates to prevent recurrence across the wider pump fleet. Team lessons learned shared at monthly reliability review. 8D closed and archived 2025-09-17.

5 WHY ROOT CAUSE ANALYSIS

Problem Statement: Mechanical seal on Pump P-xxx leaked at a rate exceeding the site acceptable limit, requiring shutdown and switch to standby pump.

Branch A : Seal Face Wear (Cavitation Path)

	Why 1	Why 2	Why 3	Why 4	Why 5
Question / Finding	Why did the mechanical seal leak? The seal faces were found scored/eroded during teardown, breaking the fluid film that normally prevents leakage.	Why were the seal faces scored/eroded? Inspection showed pitting consistent with cavitation damage, not normal wear-out.	Why was cavitation occurring at the seal? Trend data showed the pump was frequently run at flows well below its Minimum Continuous Stable Flow (MCSF) during low cooling-demand periods.	Why was the pump run below MCSF? No flow monitoring, alarm, or interlock existed to alert Operations when flow dropped below the safe minimum; operators had no visibility of the operating point relative to MCSF.	Why was there no low-flow protection or visibility? The original pump design/installation package did not include minimum-flow protection, and this gap was never flagged because low-flow risk was not included in the equipment FMEA.
Root Cause	Absence of low-flow protection (alarm/interlock) and omission of off-BEP/low-flow operation as a failure mode in the pump FMEA allowed sustained cavitating operation, which eroded the seal faces and caused leakage.				

Branch B : Installation / Alignment (Contributing Path)

	Why 1	Why 2	Why 3	Why 4	Why 5
Question / Finding	Why did the seal leak (contributing factor)? Teardown also found uneven wear across the seal face circumference, indicating a secondary contributor beyond cavitation alone.	Why was wear uneven across the face? Shaft-to-motor alignment was found outside OEM tolerance (0.15 mm TIR vs 0.05 mm spec), inducing cyclic seal face loading.	Why was alignment out of tolerance? The pump was last realigned during a coupling replacement 10 months earlier and was not re-verified afterward.	Why was realignment not re-verified? The site PM plan does not require alignment re-check after coupling work; it only mandates a check on major pump overhauls.	Why does the PM plan not cover this? The PM plan was written from a generic template and was never tailored to this pump class or updated after past coupling-related failures.
Root Cause	The preventive maintenance plan does not require alignment verification after coupling work, so misalignment introduced during the last coupling replacement went undetected and added cyclic loading that accelerated seal face wear.				

CORRECTIVE & PREVENTIVE ACTION (CAPA) TRACKER

#	Type	Action Description	Owner	Start Date	Due Date	Status	% Complete	Verification / Evidence
1	Corrective	Install low-flow switch/alarm interlock on P-xxx discharge line	Instrumentation	05/08/2025	10/08/2025	Complete	100%	Functional test record #INS-xxxx
2	Corrective	Replace mechanical seal with upgraded SiC/SiC face material	Maintenance	05/08/2025	10/08/2025	Complete	100%	Work order WO-xxxxx, teardown photos
3	Corrective	Laser-align P-xxx shaft/coupling to OEM tolerance	Maintenance	10/08/2025	10/08/2025	Complete	100%	Alignment report LA-xxxx
4	Corrective	Run 6-day post-repair validation monitoring (leak, vibration, pressure)	Reliability	11/08/2025	20256-08-17	Complete	100%	Validation log VL-Pxxx-xx
5	Preventive	Add minimum-flow interlocks to all critical centrifugal pumps (design standard update)	Reliability Eng.	18/08/2025	01/10/2025	In Progress	20%	Design standard revision - pending
6	Preventive	Update PM plan: quarterly seal chamber pressure & vibration trend review	Maintenance Planner	18/08/2025	01/09/2025	In Progress	30%	PM plan revision draft
7	Preventive	Require alignment re-verification after any coupling work (PM plan update)	Maintenance Planner	18/08/2025	01/09/2025	Not Started	0%	Pending PM plan revision
8	Preventive	Add MCSF/BEP operating window to datasheets and control room reference	Process Engineering	20/08/2025	15/09/2025	Not Started	0%	Pending
9	Preventive	Train Operations on low-flow / cavitation risk recognition	Training Coordinator	25/08/2025	15/09/2025	Not Started	0%	Training attendance sheet - pending
10	Preventive	Update Cooling Water System FMEA to include off-BEP operation as a failure mode	Quality	01/09/2025	15/10/2025	Not Started	0%	FMEA revision - pending

Summary	Overall completion:	4 / 10 actions complete	45%
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Legend: Green = Complete | Yellow = In Progress | Red = Not Started. Yellow-filled cells on other tabs indicate editable/input fields. '% Complete' and Status cells use dropdown validation and update the summary formulas automatically.